Week 9 Lecture Plan

Python GIL and Runtime Limits: Threads, Processes, and Language Models

Instructor: Dr. Mohamad Aoude

Pre-Class Preparation

☐ Week 9 Python lab runs: `week9-lab/week9-day1-python-gil-runtime-limits`

☐ Smoke test passes: `.\run-tests.ps1`

☐ Instructor check passes: `.\instructor-check.ps1`

☐ Lecture PDFs ready: `Lecture_09_01_nar.pdf` through `Lecture_09_04_nar.pdf`

☐ Lab manual ready: `Lab_Week09_PythonGilRuntimeLimits.pdf`

☐ Open visual proof tool: Task Manager, `htop`, or `py-spy`

Learning Targets

By the end of Week 9, students must be able to:

1. Explain why CPU-bound CPython threads do not scale across cores.
2. Compare CPU-bound Python threads with multiprocessing.
3. Capture visual evidence of runtime behavior.
4. Explain multiprocessing overhead.
5. Contrast Python threads with Go goroutines and Erlang actors.
6. Connect Knight Capital to timing, deployment, and operational control risk.

Lecture Structure

Segment	Material	Goal
Part 1	`Lecture_09_01_nar`	GIL reality and CPU-bound threads
Part 2	`Lecture_09_02_nar`	Multiprocessing and visual proof
Part 3	`Lecture_09_03_nar`	Go goroutines and Erlang actors contrast
Part 4	`Lecture_09_04_nar`	Knight Capital discussion and Week 10 bridge

Recommended Time Budget

Time	Activity	Notes
0–10 min	Week 8 handoff	Debugging habit: measure before claiming speed.
10–40 min	Part 1 slides and Demo01	GIL, CPU-bound threads, workload specificity.
40–70 min	Part 2 slides and visual proof	Threads vs processes, Task Manager or htop.
70–90 min	Part 3 slides and Demo03	Go/Erlang contrast.
90–110 min	Part 4 discussion	Knight Capital operational failure.
110–120 min	Exit ticket	Runtime model, evidence, tradeoff.

Exit Ticket

1. Why do CPU-bound CPython threads fail to scale across cores?
2. What cost does multiprocessing introduce?
3. What concurrency lesson does Knight Capital reinforce?